Qianxun (Cecilia) Xu

+1 323-202-5376 · qianxun45@ucla.edu · https://ceciliathebirb.github.io

EDUCATION

University of California, Los Angeles 2026–2028
M.S. in Computer Science

Duke Kunshan University & Duke University Dual Degree 2022-2026

- B.S. in Applied Mathematics and Computer Science; Computer Science Track
- GPA: 3.93/4.0 | Major GPA: 3.97/4.0
- Awards: IEEE ICPADS Best Presentation in Session; NeurIPS Scholar Award ($2,000); DKU Merit-based College Entrance Scholarship ($48,000); Graduated Summa Cum Laude (Top 5%).

PUBLICATIONS

Xu, Q., Song, C., Cai, Y., Zhang, C. SwitchCraft: Training-Free Multi-Event Video Generation with Attention Controls. ***CVPR****, 2026 (main track poster).*

Xu, Q., Boukerche, A., Sun, P. FENSe: Feedback-Enabled Neighbor Selection for Spatial Aware Collaborative Perception. ***IEEE ICPADS****, 2025 (oral).*

Xu, Q., Li, Z. Partial Physics Informed Diffusion Model for Ocean Chlorophyll Concentration Reconstruction. ***NeurIPS****, 2025 (main track poster).*

INDUSTRY EXPERIENCE

StepFun | Research Intern, World Models Apr–Sep 2026
ConsistWorld: Evidence Routing for Consistent Multi-Agent World Models

- Designed memory retrieval and visibility-gated sharing with bounded active context, reducing cross-agent appearance inconsistency by **69%** on asynchronous handoffs versus a baseline without either module.
- Built an end-to-end **synthetic-data pipeline** integrating calibrated camera geometry and motion-constrained trajectory generation to produce aligned multi-view videos with diverse revisit patterns for training consistent world models.
- Developed a **staged training curriculum**: stable teacher-forced initialization on general multi-view data, self-resampling to mitigate rollout errors, and targeted adaptation on synthetic revisit data.
- Designed **consistency evaluation protocols** for revisits and cross-agent sharing, validating generalization to unseen agent counts and longer rollouts. Demonstrated improved visual fidelity, reducing perceptual error by **19%** compared with full-history attention.

RESEARCH EXPERIENCE

UCLA | Graduate Researcher Sep 2026–Present
Efficient Sidewalk World Model Distillation | Advisor: Bolei Zhou

- Adapted **AnyFlow-based distillation** for low-latency world models with a **non-uniform denoising schedule** and **noisy KV reuse**, aligning training rollouts and history supervision with inference to reduce redundant computation.
- Achieved better throughput (**1.61**× FPS) and better visual quality (**4.3%** lower LPIPS) on 200 validation clips versus the base AnyFlow student.
- Built an **automated training–evaluation workflow** for multi-GPU distillation, enabling systematic comparison of sampling schedules and cache policies through reproducible visual-quality and inference-throughput benchmarks.

Westlake University | Undergraduate Researcher May–Nov 2025
SwitchCraft: Training-Free Multi-Event Video Generation | Advisor: Chi Zhang

- Implemented **Event-Aligned Query Steering** in Wan2.1 T2V-14B cross-attention, projecting frame-specific queries toward target-event subspaces and away from competing events to control event timing without fine-tuning.
- Formulated adaptive steering as a **convex optimization problem**, selecting enhancement and suppression strengths from attention margins to prevent oversteering and preserve appearance without prompt-specific tuning.
- **Results:** Achieved SOTA text-to-video alignment among evaluated methods, improving alignment scores by **24%** over the Wan2.1 baseline while preserving temporal smoothness and visual quality.

Duke Kunshan University | Undergraduate Thesis Jan–Aug 2025
Communication-Efficient Collaborative Perception | Advisor: Peng Sun

- Developed **feedback-guided neighbor selection** for bandwidth-constrained V2X perception, using **traffic-aware clustering** to eliminate redundant feature transmission.
- **Results:** Achieved superior accuracy-efficiency trade-off. Improved detection accuracy by **~4%** at matched bandwidth while reducing ego-side computation.

Duke Kunshan University | Undergraduate Researcher May 2024–May 2025
Partial Physics-Informed Diffusion Model | Advisor: Zuchuan Li

- Developed a physics informed diffusion model for sparse ocean reconstruction, combining **selective physics encoding** and **subspace alignment** to incorporate partially known dynamics.
- **Results:** Achieved robust long-range spatiotemporal inference. Reduced reconstruction RMSE by **34%** against pure data-driven models, and by **49%** against baselines forcing complete physical equations.

SKILLS & LANGUAGES

Technical: Python, Java, C, PyTorch, Git, Linux, distributed training
Languages: Mandarin (native), English (TOEFL 116/120), German (intermediate)